

TESTING ENGINEERING FUNDAMENTALS

Software Engineering Seminar

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

1 Basic Concepts

2 Testing Levels



Outline

1 Basic Concepts

2 Testing Levels



Software Engineering

- **Software engineering** is the application of engineering principles to the design, development, and maintenance of software systems.
- It involves the use of systematic methods and tools to ensure that software is of high quality, reliable, and meets the needs of its users.
- The main goal of software engineering is to produce software that is cost-effective, efficient, and maintainable.
- It encompasses a wide range of activities, including requirements analysis, design, implementation, testing and maintenance.
- Software engineering is a collaborative discipline that involves collaboration between developers.



Software Testing

- **Software testing** is the process of **evaluating** a software **system** or its **components** to determine whether it **satisfies** the specified **requirements** and to **identify any defects**.

[Functional

non-functional $\Rightarrow$ quality

Fix before user detect them



Software Testing

- **Software testing** is the process of **evaluating** a software system or its components to determine whether it **satisfies** the specified **requirements** and to **identify any defects**.
- It involves **executing the software under controlled conditions** and evaluating the **results against expected outcomes**.
- The *main goal of software testing* is to ensure that the software is of high quality, reliable, and meets the needs of its users.
- It is an **essential** part of the **software development process** and is typically performed by independent testers or quality assurance teams.

test
design

Users
Infrastructure
resources



Software Testing

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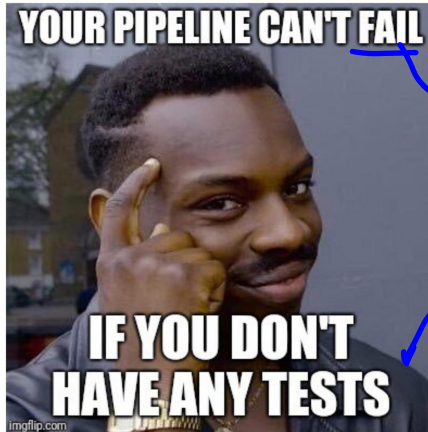
Software developers
↓
Unit test



Classical Philosophy

"If you don't have time to do it right, when will you have time to do it over?"

↳ project management



Software Quality

- **Software quality** is the degree to which a software system meets the specified requirements and satisfies the needs of its users.
- It is a critical aspect of software development and is typically measured by a set of **quality attributes**, such as **reliability**, **performance**, **usability** and **maintainability**.
- **Software quality** is typically assessed through a combination of **static analysis**, **dynamic analysis**, and **testing** techniques.

Handwritten notes and arrows:

- A blue arrow points from "degree" to "meets".
- A red arrow points from "quality attributes" to "reliability".
- A red arrow points from "reliability" to "multiple experiments".
- A red arrow points from "multiple experiments" to "dynamic analysis".
- A red arrow points from "dynamic analysis" to "generic".
- Handwritten text: "generic", "dynamic", "multiple experiments", and a vertical list of symbols: "10", "18", ":", "7", "~", "~", "~", "~".



Quality is a Key Factor

**“No pasa
nada, así
mándalo a
producción”
by
Crowdstrike**

Figure: Microsoft Style!



Testing Engineering

- **Testing engineering** is the application of engineering principles to the **design**, **development**, and **execution** of **tests** for software systems
- It involves the use of **systematic methods** and **tools** to ensure that software is of **high quality**, **reliable**, and **meets the needs of its users**.
- The main goal of testing engineering is to produce software that is cost-effective, efficient, and maintainable through the use of automated testing and test-driven development (TDD) practices.
- It encompasses a wide range of activities, including test design, test execution, and test analysis.
- Testing engineering is a collaborative discipline that involves collaboration between developers and testers to ensure that software is of high quality and meets the needs of its users.

static code
 ↓
 compilation error

steps

execution time
 ↓
 common



Testing Engineering

- $11q + j \Rightarrow 11q$
- $C \rightarrow \text{Sum}(2, n2) \rightarrow 9999999, 99999999$
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 - It involves the use of **systematic methods** and **tools** to ensure that software is of **high quality**, **reliable**, and **meets the needs** of its users.
 - The **main goal** of testing engineering is to produce **software** that is **cost-effective**, **efficient**, and **maintainable** through the use of **automated testing** and **test-driven development** (TDD) practices.

↓
save money
but
study

↓
design the test
before code
→ edge cases

↓
happy path X



Testing Engineering

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- It encompasses a wide range of activities, including **test design**, **test execution**, and **test analysis**.
- **Testing engineering** is a **collaborative** discipline that involves **collaboration** between **developers** and **testers** to ensure that software is of **high quality** and **meets the needs of its users**.

test plan

product owner



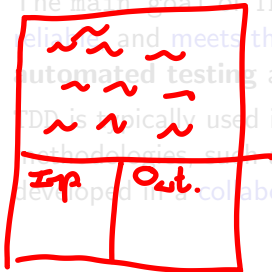
Test-Driven Development (TDD)

~ last 10 years

- **Test-Driven Development (TDD)** is a *software development methodology* that emphasizes the use of **automated tests** to drive the design and development of software.
- It involves **writing a test** for a specific piece of functionality **before** writing the code to implement that functionality.

- The main goal of TDD is to produce software that is of high quality, reliable and meets the needs of its users through the use of automated testing and test-driven development practices.
- TDD is typically used in conjunction with other software development methodologies, such as Agile or Scrum, to ensure that software is developed in a collaborative and iterative manner.

* Unit tests



programming
contest



Test-Driven Development (TDD)

- **Test-Driven Development (TDD)** is a *software development methodology* that emphasizes the use of **automated tests** to drive the design and development of software.
- It involves **writing a test** for a specific piece of functionality **before** writing the code to implement that functionality.
- The main goal of **TDD** is to produce software that is of **high quality**, **reliable**, and **meets the needs of its users** through the use of **automated testing** and **test-driven development** practices.
- TDD is typically used in *conjunction* with other software development methodologies, such as (*Agile* or *Scrum*), to ensure that software is developed in a **collaborative** and **iterative** manner.



Quality Guidelines

- **Quality guidelines** are principles that guide the design of a system to ensure that it meets the needs of its users.
- They include reliability, scalability, maintainability, and usability guidelines.
- They are *important* for *ensuring* that a **system** is robust, efficient, and effective.

~ 30
great product — sells — stable



Reliability Guidelines

- **Reliability guidelines** are *principles* that *guide* the *design* of a *system* to *ensure* that it is *reliable* and *dependable*.
- They include *fault-tolerance*, *redundancy*, and *error-handling* guidelines.
- They are *important* for *ensuring* that a *system* is *robust* and *resilient* to *failures*.



Scalability Guidelines

- **Scalability guidelines** are *principles* that *guide* the *design* of a system to *ensure* that it is *scalable* and *flexible*.
- They include modularity, extensibility, and performance guidelines.
- They are *important* for *ensuring* that a *system* can *grow* and *adapt* to changing requirements.

→ microservices

auto scaling



Maintainability Guidelines

- **Maintainability guidelines** are *principles* that *guide* the *design* of a *system* to *ensure* that it is *easy* to *maintain* and *update*.
- They include *modularity*, *documentation*, and *versioning* guidelines.
- They are *important* for *ensuring* that a *system* can be *easily* *maintained* and *updated* by its *developers*.



Quality Standards

- **Quality standards** are **benchmarks** that *define* the **level** of **quality** that a **system** must **meet**.
- They include **ISO 9000**, **CMMI**, and **Six Sigma** standards.

german

9000 - Focus Cpe



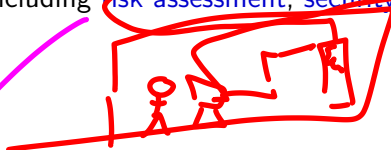
ISO 9000

- **ISO 9000** is a quality standard that *defines* the **requirements** for a **quality management system**.
- It is *designed* to help **organizations** ensure that they **meet** the **needs** of their **customers** and **stakeholders**.
- It is *based* on a **number of quality management principles**, including **customer focus**, **leadership**, and **continuous improvement**.



ISO 27001

- **ISO 27001** is a quality standard that *defines* the **requirements** for an **information security management system**.
- It is *designed* to help **organizations** protect their **information** and **ensure** that it is **secure** and **confidential**.
- It is *based* on a number of information security management principles, including **risk assessment**, **security policies**, and **incident response**.



→ risk assessment



CMMI → *Capitilad ⇨ Enterprise*

- **CMMI** is a quality standard that *defines* the **requirements** for a mature software development process. *8-5 ⇒ min 3*
- It is *designed* to help organizations improve their software development processes and deliver high-quality products to their customers.
- It is *based* on a number of best practices for software development, including *requirements management*, *project planning*, and *process monitoring*.

→ mature process



Six Sigma

- **Six Sigma** is a quality standard that *defines* the requirements for a process that is *capable* of producing high-quality products.
- It is *designed* to help organizations improve their processes and reduce defects in their products and services.
- It is *based* on a number of quality management principles, including data-driven decision-making, process improvement, and customer focus.

Plan



Outline

1 Basic Concepts

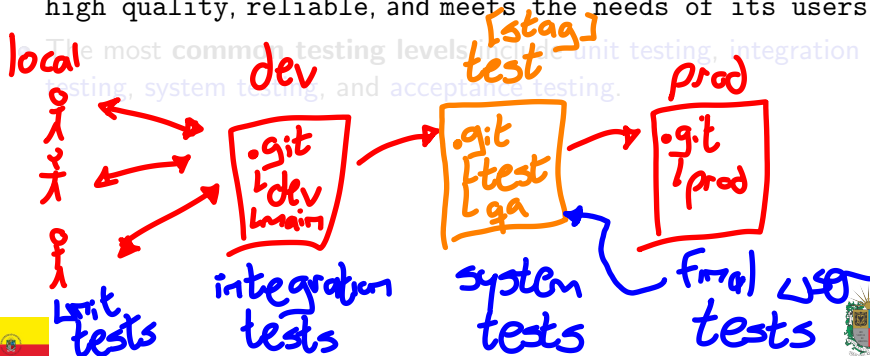
2 Testing Levels



Testing Levels

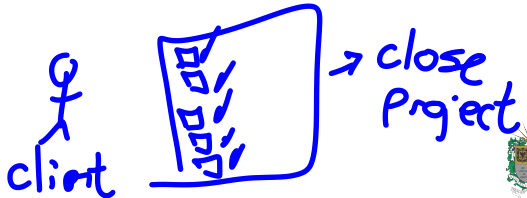
Dev-001 → feature/dev-001
bug/dev-001

- **Testing levels** refer to the different stages of testing that a software system goes through during its development lifecycle.
- Each level of testing has its own set of objectives, techniques, and tools.
- The **main goal** of testing levels is to ensure that the software is of high quality, reliable, and meets the needs of its users.



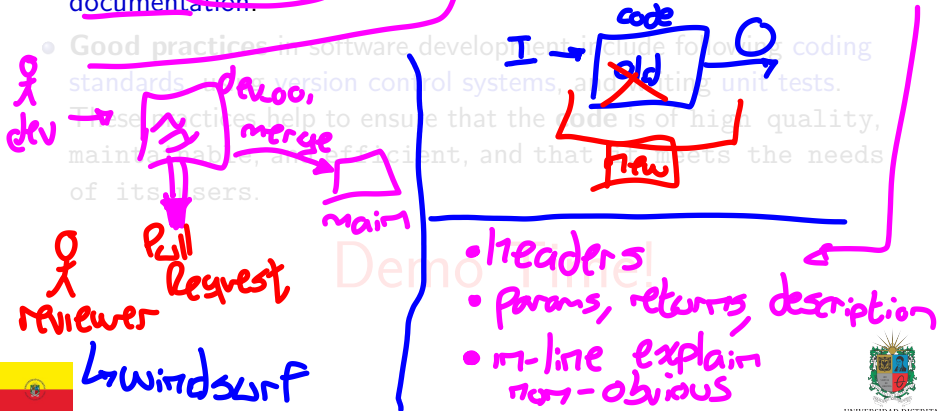
Testing Levels

- **Testing levels** refer to the **different stages** of testing that a software system goes through during its **development lifecycle**.
- **Each level** of testing has its own set of **objectives**, **techniques**, and **tools**.
- The **main goal** of testing levels is to **ensure** that the software is of high quality, reliable, and meets the needs of its users.
- The most **common testing levels** include unit testing, integration testing, system testing, and acceptance testing.



Base Code Quality & Good Practices

- **Base code quality** refers to the fundamental principles and practices that ensure that the **code** is of high quality, maintainable, and efficient.
- It includes **practices** such as code reviews, code refactoring, and code documentation.



Base Code Quality & Good Practices

- **Base code quality** refers to the **fundamental principles** and **practices** that ensure that the **code** is of high quality, maintainable, and efficient.
 - It includes **practices** such as **code reviews**, **code refactoring**, and **code documentation**.
 - **Good practices** in software development include following **coding standards**, using **version control systems**, and writing **unit tests**.
- These practices help to ensure that the **code** is of high quality, maintainable, and efficient, and that it meets the needs of its users.

PEP 8

Python snake

Java - Camel

Docs → Google

Demo Time!

Git tags?
changes
control?



Static Analysis

- **Static analysis** is a *software testing technique* that involves *analyzing the source code* of a software system *without executing it*.
- The **main goal** of static analysis is to identify *potential defects*, *vulnerabilities*, and *code quality issues* in the source code.
- **Static analysis tools** can automatically analyze the source code and provide feedback on potential issues, such as *coding standards violations*, *security vulnerabilities*, and *performance issues*.
- *Static analysis* is typically performed *early* in the software development process.

Demo Time!



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Demo Time!

PyLint



Unit Testing

- **Unit testing** is a software testing technique that involves testing individual units or components of a software system in isolation.
- The main goal of unit testing is to identify defects and issues in the source code at an early stage of the software development process.
- **Unit tests** are typically written by developers and are executed automatically as part of the software development process.
- **Unit testing** is typically performed after the code has been written and before the code is integrated into the larger software system.
- **Unit testing** is an essential part of the software development process and is typically performed using unit testing frameworks such as JUnit, NUnit, or pytest.

Demo Time!



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- ↓
avoid bad code
in main branch



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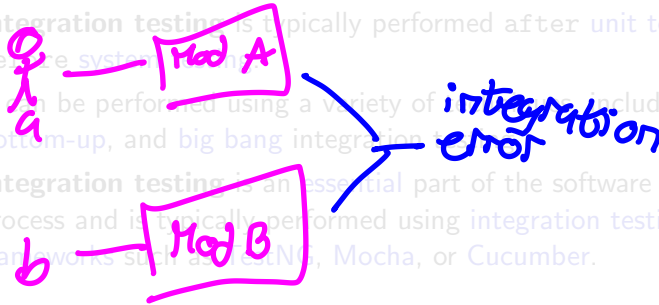
JUnit NUnit Python
 Java .NET Demo Time!



Integration Testing

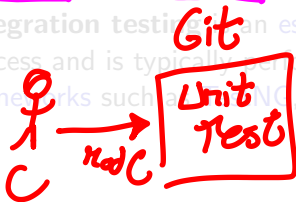
- **Integration testing** is a *software testing technique* that involves testing the interactions between different components or units of a software system.
- The main goal of integration testing is to identify defects and issues in the interactions between different components or units of the software system.

- Integration testing is typically performed after unit testing and before system testing.
- It can be performed using a variety of approaches including top-down, bottom-up, and big bang integration testing.
- Integration testing is an essential part of the software development process and is typically performed using integration testing frameworks such as TestNG, Mocha, or Cucumber.



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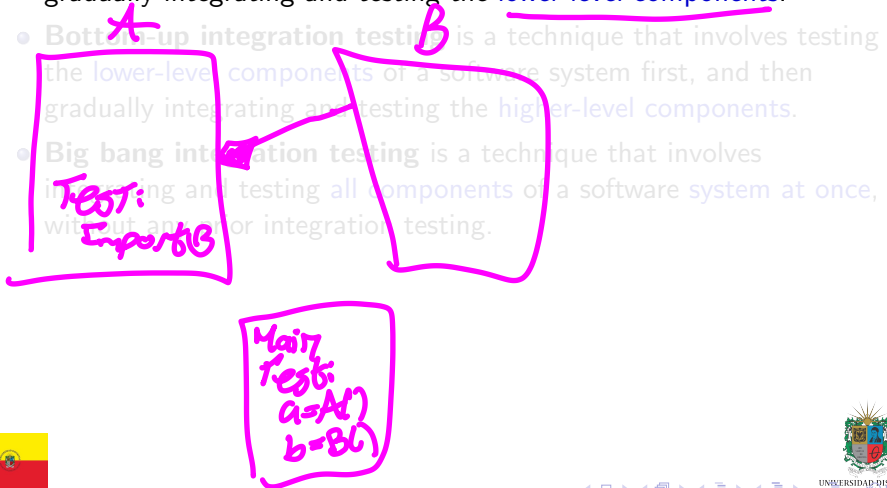


Apache



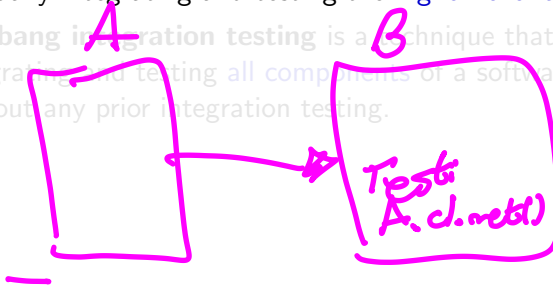
Integration Testing Techniques

- **Top-down integration testing** is a technique that involves testing the higher-level components of a software system first, and then gradually integrating and testing the lower-level components.



Integration Testing Techniques

- **Top-down integration testing** is a technique that involves testing the **higher-level components** of a software system first, and then gradually integrating and testing the **lower-level components**.
- **Bottom-up integration testing** is a technique that involves testing the **lower-level components** of a software system first, and then gradually integrating and testing the **higher-level components**.
- **Big bang integration testing** is a technique that involves integrating and testing all components of a software system at once, without any prior integration testing.



Integration Testing Techniques

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- **Big bang integration testing** is a technique that involves integrating and testing **all components** of a software system at once, without any prior integration testing.

Fork



API Testing → web services

- **API testing** is a software testing technique that involves testing the application programming interfaces (APIs) of a software system.
- The **main goal** of API testing is to identify defects and issues in the APIs of the software system, including their functionality, performance, and usability.
- **API testing** is typically performed after integration testing and before system testing.
- It can be performed using a variety of techniques, including functional, non-functional, and regression testing.
- **API testing** is an essential part of the software development process and is typically performed using API testing frameworks such as Postman, SoapUI, or RestAssured.

main.py / subapi1.py
subapi2.py
Integration Test

Demo Time!



API Testing

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- **API testing** is typically performed after **(integration testing)** and before **(system testing.)**
- It can be performed using a variety of **techniques**, including **functional**, **non-functional**, and **regression** testing.

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Handwritten notes:

- Red arrows pointing from "functional" and "non-functional" to "input vs output".
- Red arrow pointing from "regression" to "performance".
- Blue circle around "regression".
- Blue text: "add test old code is working" with a box around "add".

Demo Time!



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Demo Time!



System Testing

- **System testing** is a *software testing technique* that involves testing the entire software system as a whole. *→ MVP*
- The **main goal** of system testing is to identify defects and issues in the software system as a whole, including its functionality, performance, and usability.
- **System testing** is typically performed after integration testing and before acceptance testing.
- It can be performed using a variety of **techniques**, including functional, non-functional, and regression testing.
- **System testing** is an essential part of the software development process and is typically performed using system testing frameworks such as Selenium, Appium, or Robot Framework.



System Testing

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- **System testing** is typically performed after integration testing and before acceptance testing.
- It can be performed using a variety of ~~techniques~~, including functional non-functional, and regression testing.
- *User stories* → System testing is an essential part of the software development process and is typically performed using system testing frameworks such as Selenium, Appium, or Robot Framework.



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System Testing Techniques

- **Functional testing** is a technique that involves testing the functionality of a software system to ensure that it meets the specified requirements.
 - **Non-functional testing** is a technique that involves testing the non-functional aspects of a software system, such as its performance, usability, and security.
 - **Regression testing** is a technique that involves retesting a software system after changes have been made to ensure that the changes have not introduced any new defects.
- Handwritten notes:*
- A red box highlights "specified requirements." with an arrow pointing to "Usability" under Non-functional testing.
 - A red arrow points from "functionality" to "Functional" under Non-functional testing.
 - A blue bracket under "Regression testing" is labeled "acceptance criteria".
 - Red handwritten text "stories" is written near the "Regression testing" bullet.



System Testing Techniques

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Product Quality

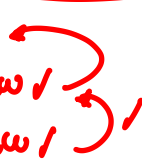
UI/UX



System Testing Techniques

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1.0 /
1.1 new /
1.2 new /



Stress Testing

- **Stress testing** is a *software testing technique* that involves **testing the software system under extreme conditions** to determine its **stability** and **performance**.
- The **main goal** of stress testing is to identify **defects and issues** in the software system when it is subjected to **high loads** or **stressful conditions**.
- **Stress testing** is typically performed after **system testing** and before **acceptance testing**.
- It can be performed using a variety of **techniques**, including **load testing**, **performance testing**, and **scalability testing**.
- **Stress testing** is an **essential** part of the software development process and is typically performed using **stress testing frameworks** such as **JMeter**, **Gatling**, or **Locust**.

Demo Time!



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Demo Time!



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Demo Time!



Acceptance Testing

- **Acceptance testing** is a *software testing technique* that involves testing the software system to ensure that it meets the needs of its users. *user stories*
 - The **main goal** of acceptance testing is to identify **defects and issues** in the software system **from the perspective of the end user**.
 - Acceptance testing is typically performed after system testing and before deployment.
 - It can be performed using a variety of **techniques**, including user acceptance, alpha, and beta testing.
 - Acceptance testing is an essential part of the software development process and is typically performed using acceptance testing frameworks such as Cucumber, FitNesse, or Behave.
- close the project*
- Acceptance criteria**

Demo Time!



Acceptance Testing

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test / QA
environment

Demo Time!



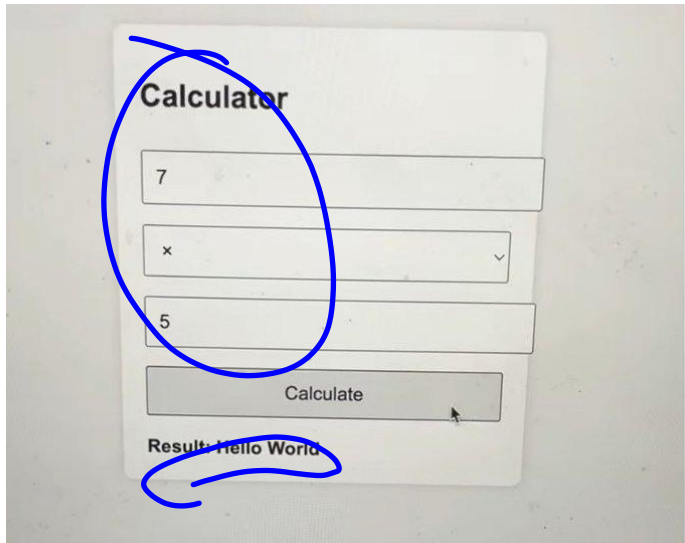
Acceptance Testing

- **Acceptance testing** is a *software testing technique* that involves testing the software system to ensure that it meets the needs of its users.
- The **main goal** of acceptance testing is to identify **defects** and **issues** in the software system from the perspective of the end user.
- **Acceptance testing** is typically performed after **system testing** and *before deployment*.
- It can be performed using a variety of **techniques**, including **user acceptance**, **alpha**, and **beta** testing.
- **Acceptance testing** is an **essential** part of the software development process and is typically performed using **acceptance testing frameworks** such as **Cucumber**, **FitNesse**, or **Behave**.

Demo Time!



Feature Error



Outline

1 Basic Concepts

2 Testing Levels



Thanks!

Questions?



Repo: www.github.com/EngAndres/ud-public/tree/main/courses/software_engineering_seminar

